

How Can a Wellness Technology Company Play It Smart?

Ask:

Business Task: What are trends in smart device usage and how can these trends be applied to Bellabeat customers and how can they influence Bellabeat marketing strategies?

Prepare:

The data used was provided through a Kaggle dataset, it is FitBit Fitness Tracker Data. It includes data from 35 users from March and April of 2016.

Process:

SQL was used for this analysis as it was determined that some of the data would exceed the capacity of Microsoft Excel. Initial exploration revealed that 138 out of 1,397 rows contained zero steps. Further investigation showed that only 9 rows had zero values across all activity metrics (including 1,440 sedentary minutes), indicating that the device was likely not worn on those days. These rows were excluded from the analysis, while other zero-step entries were retained as they may represent valid low-activity days.

Analyze:

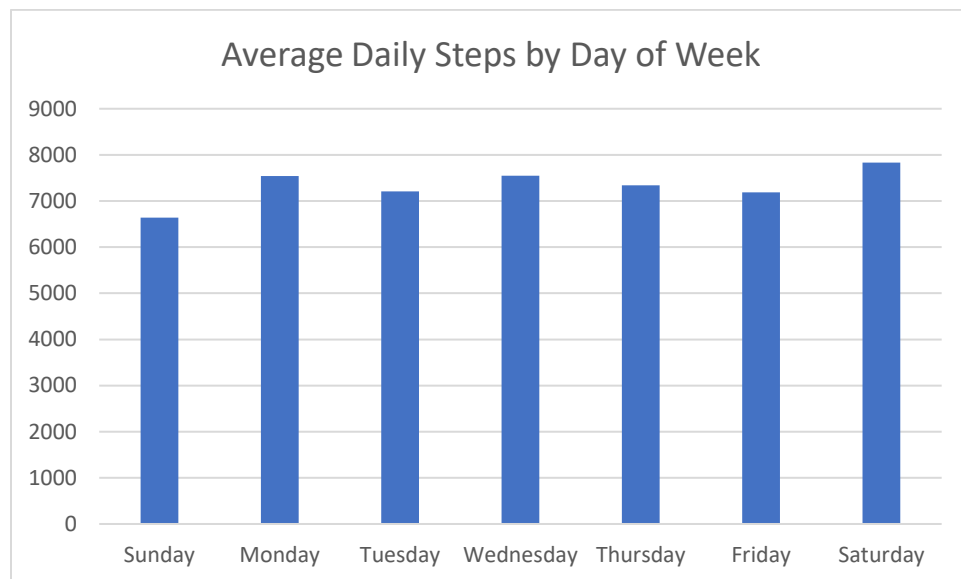
For the analysis average daily steps, and average calorie consumption by day of the week was calculated. Using average daily step counts, users were segmented into low-, moderate-, or high-activity groups based on whether they averaged fewer than 5,000, between 5,000 and 10,000, or more than 10,000 average daily steps. A distribution of active minutes by activity intensity was also calculated, broken down into either lightly active, fairly active, or very active. These are the main findings from the analysis:

- Users were most active on Saturday, with average step counts peaking compared to all other days of the week, and least active on Sunday, the day of the week with the lowest average step count.
- User activity levels were mostly consistent across weekdays.
- Calories burned were relatively consistent across the full week, suggesting that step count may be a more useful metric for identifying behavioral differences.
- 80% of users had an average daily step count of less than 10,000, falling into the low- or moderate-activity levels, while only 20% qualified as highly active users.

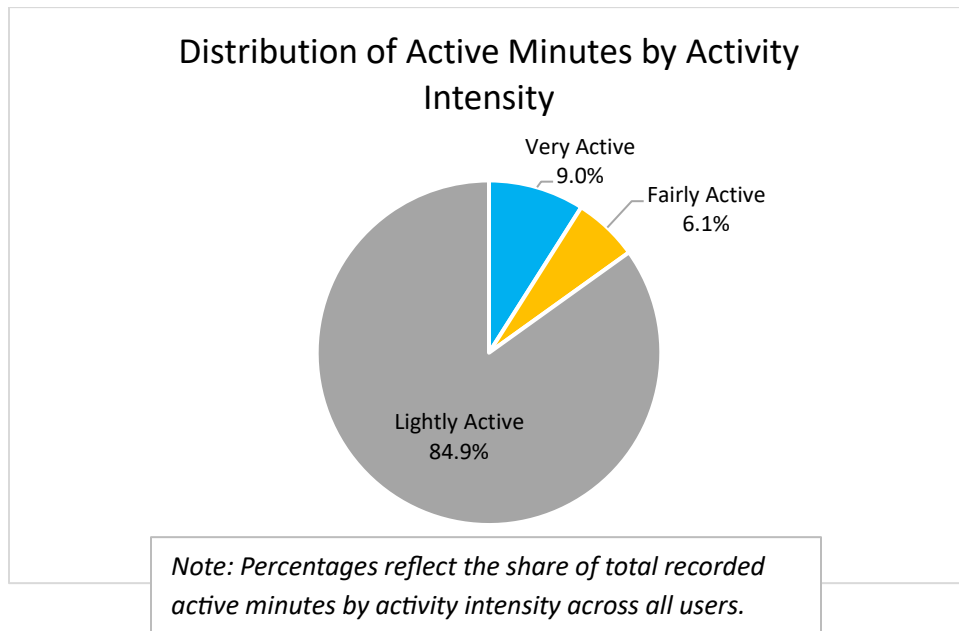
- Lightly active minutes accounted for 84.9% of all recorded active minutes, while fairly active and very active minutes accounted for 6.1% and 9.0% respectively.

Share:

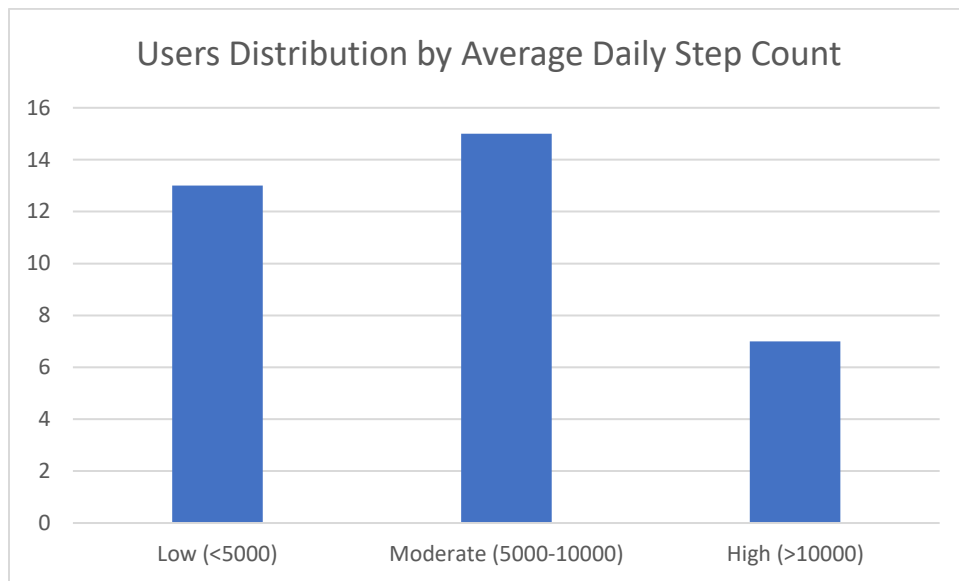
This chart shows the average daily steps taken by users based on the day of the week. Saturday step count is highest and Sunday step count is lowest suggesting that there is an opportunity to amplify users Saturday activity levels and offer engagement to strengthen Sunday levels. User activity levels on weekdays are relatively consistent, suggesting that users have standard routines during the weekdays that could be reinforced.



Next, this chart shows the distribution of active minutes by intensity level. The majority of recorded active minutes were light-intensity, suggesting users engage more frequently in low-intensity movement than sustained moderate or vigorous exercise. This suggests an opportunity to direct marketing efforts towards indicating the wearability of a tracker during all hours of the day, not just during rigorous exercise.



Finally, this chart shows the number of users segmented into groups based on their average daily step count. 13 of the 35 users (37.1%) had fewer than 5,000 average daily steps, these users were placed in the low-activity category. 15 (42.9%) had between 5,000 and 10,000 steps placing them in the moderate-activity category, and the remaining 7 users (20%) had more than 10,000 average daily steps, these users were considered highly active and placed in the corresponding category. By this metric 80% of users were classified as low- or moderate-activity users, suggesting that Bellabeat may benefit from prioritizing low- and moderate-activity users, who represent approximately 80% of the sample.



Act:

Overall, the analysis suggests that users engage primarily in low-intensity activity, maintain relatively consistent weekday routines, and are predominantly low- to moderate-activity users. With that in mind, the following recommendations are based on the user generated FitBit data, with the goal of applying the analysis to Bellabeat customers and any future Bellabeat marketing strategies:

1. Bellabeat could aim to engage users on Sunday (the day of the week with the fewest daily steps) by offering incentives and/or reminders to engage in low-intensity activities such as going on a casual walk or a light bike ride.
2. Bellabeat could look to further amplify strong Saturday levels by offering challenges to meet a certain activity goal on that day of the week, or on the weekend in general to also help with Sunday levels.
3. Bellabeat could engage in marketing that emphasizes the wearability and comfort of their product, emphasizing that it can be used across all activity levels, not only for intense physical activity.

Limitations:

- The dataset used is from 2016, so it may not fully reflect current trends
- The dataset includes a relatively small sample size (35 users), which may limit generalization
- Data may include inconsistencies (e.g. device not worn)
- Sample may not represent all smart device users

Conclusion:

The analysis found that users were most active on Saturdays and least active on Sundays, spent the majority of active minutes in low-intensity activity, and were predominantly low- to moderate-activity users. These findings suggest that Bellabeat's marketing efforts may be most effective when focused on habit-building, accessibility, and weekend engagement. While the dataset is limited by its age and sample size, it provides useful insight into wearable-device user behavior.